EPSR-test-ex107 ref_intra (EP and reference potential inside molecules) + control-p bc

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the `ref_intra` keyword of the `epsr` engine [1,2,3], which lets a fraction of the empirical potential and of the full reference potential act on the intramolecular structure, combined with pressure control along the b and c axes only.

Code exercised

Fortran (original, fortran_src/):

- `molfit.f`: `en_atom_intra` (adds `refine_intra_fac` × EP to bonded and non-bonded intramolecular pairs)
- `epsr_standalone_rev3.f`: `generate_rp` (`xav` = `coul_intra_fac*ecoul`), `runepsr17` (`ref_intra`, `control-p`)
- `ato_manipulation_rev4.f`: `change_boxsize`, `press`

Rust (port, rust/epsr-core/src/):

- `energy.rs`: `en_atom_intra`; `potential.rs`: `generate_rp`
- `pressure.rs`: `control_pressure`, `change_boxsize`; `energy.rs`: `press`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. With `ref_intra 0.5 0.3` the intramolecular energy of bonded pairs (harmonic) and non-bonded pairs gains 0.5 × the empirical potential, and the intramolecular potential of every pair is replaced by 0.3 × the full reference potential (LJ + Coulomb + Gaussians + repulsive core). `control-p bc` scales b and c by `(1+rat)**0.5`, so the box becomes orthorhombic.

Inputs

`scripts/make_inputs.sh`: `.EPSR.inp` with `ref_intra 0.5 0.3`, `control-p bc 0.0 0.01`, `ntimes 2`, `niter 2`. `commands.txt`: `epsr 200DMSO400Water.EPSR.inp`, `repeat 2`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
The README reports gcov counts of `molfit.f`: the bonded `refine_intra` line executes 51600 times, the non-bonded one 37800 times, and `xav(j,ic)=coul_intra_fac*ecoul(j,ic)` 69678 times. The stored Fortran reference `.ato` has a = 33.1213, b = c = 33.6477 Å after four iterations, written with the general-cell header and re-read in pass 2.

Notes

Reproduced: `xav(npt)` is still zeroed after the replacement and the gradient `xav1` is computed before that, as in the Fortran.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.